Towards a Digital Twin of Ottawa's Transportation System

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Introduction

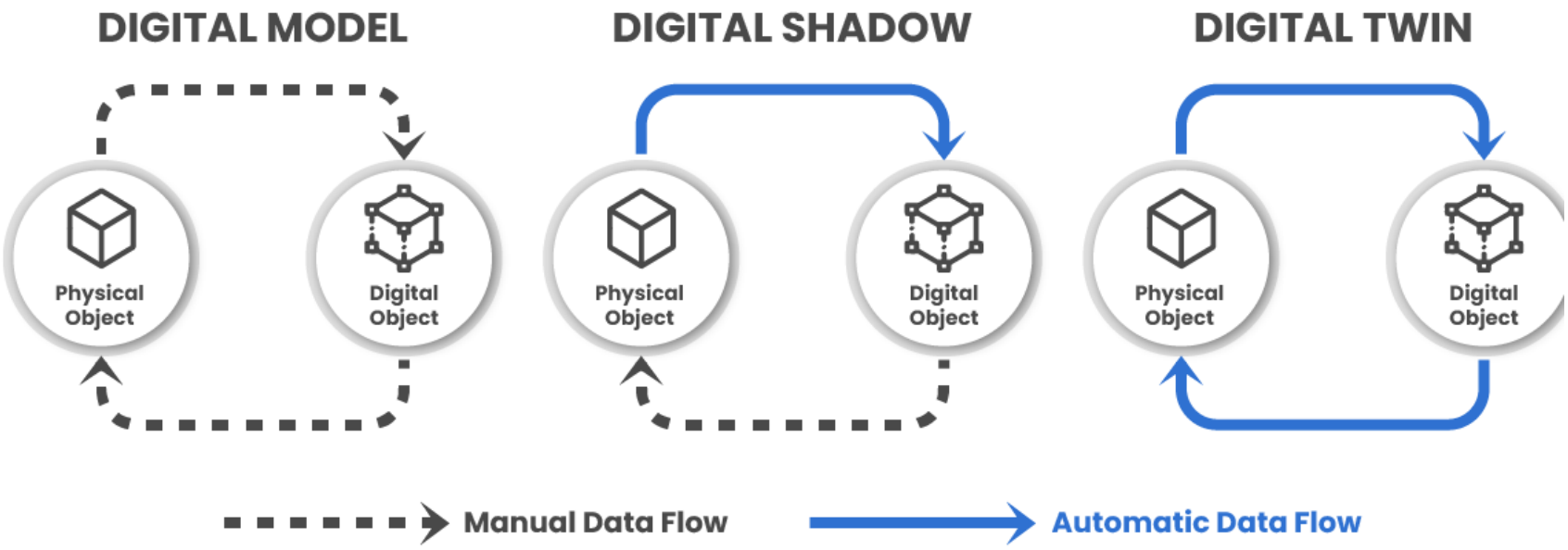
- Need to **reduce**:
  - Travel time
  - Operation costs
  - CO2 emissions
- Need for **urban planning tools**
- New Ottawa Arena
- Can Ottawa's transportation system **cope**?
- Plans to **adapt**?
- Plans for **evacuation**?

https://www.cbc.ca/news/canada/ottawa/ottawa-sens-lebreton-flats-1.7652663

Introduction

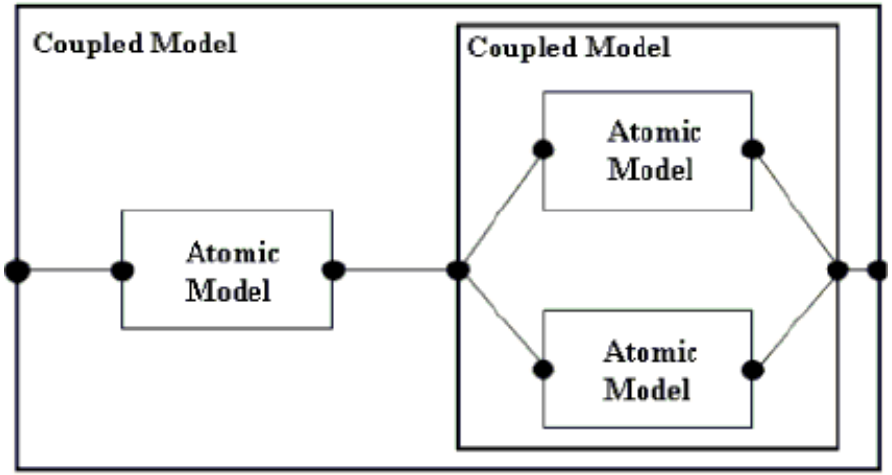
- **Digital Twins,** e.g., Smart Open Hamburg
- Digital Twin is too complex…
- New Ottawa Arena is too complex…
- Project Scope:
  - **Foundation for a Digital Twin** of Ottawa's transportation system
  - **Digital Model** designed for **reuse** and **extensibility**
  - Demonstrate the system with a **model and simulation** of a **university campus evacuation**

Background: Digital Twin



Background: DEVS Formalism

- Modeling language
- Hierarchical
- Discrete-event
- Atomic vs. Coupled Models
- Modular Components



Background: Diffusion Processes & OD data

- **Diffusion** is the process of an object **spreading** throughout a system
- **Origin-destination (OD) data** is the **flow rate** of an object from one place to another.
- Used to configure **diffusion behaviour**
- Used in **traffic simulations**

Experimental Frame

Scenario:

- Vehicle only **evacuation** of Carleton campus
- Vehicles leave **Parking Lots** through **Intersections** and **Roads** until they have **evacuated** the campus.
- **Assumptions:**
  - All parking lots are **full**
  - OD data is **consistent** across all runs
- **Scenario 1:** Vehicles begin leaving all Parking Lots at immediately.
- **Scenario 2:** Vehicles begin leaving P1 .. P4 immediately, Vehicles begin leaving P5 .. P7 after one hour.
- **Scenario 3: …**

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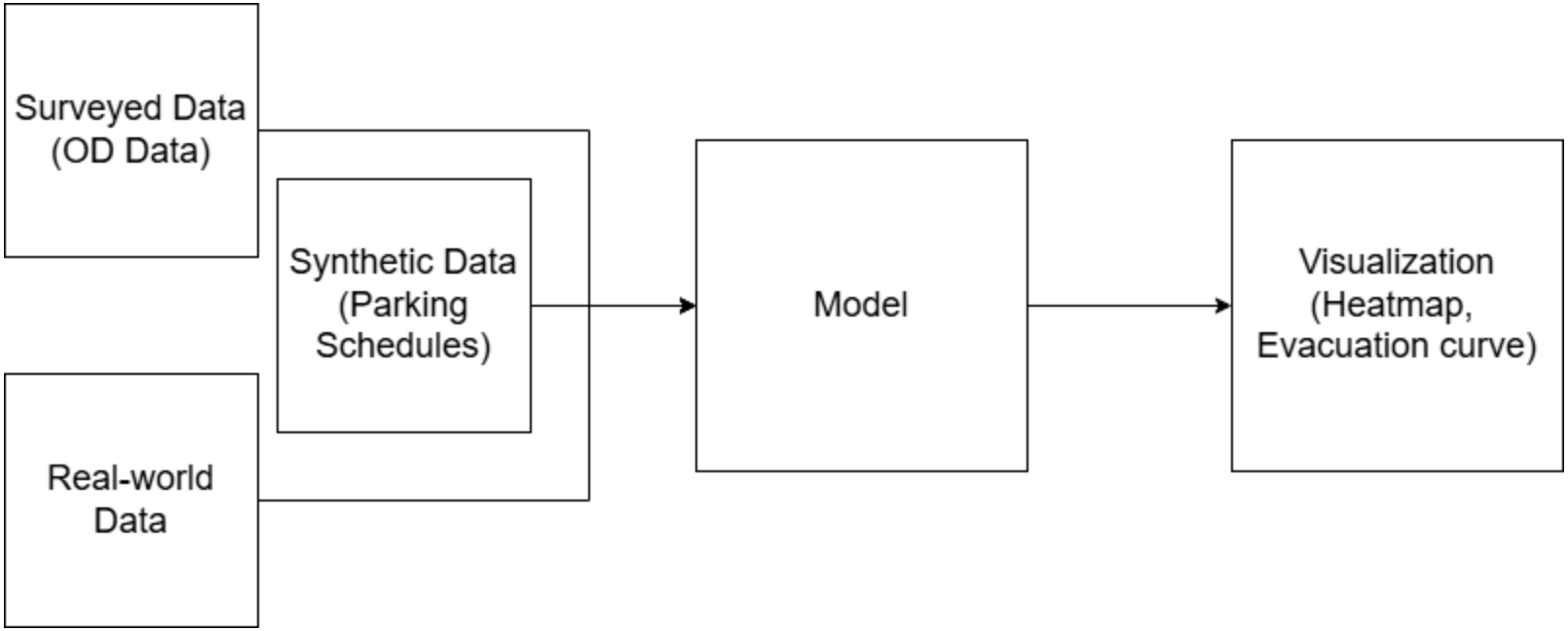
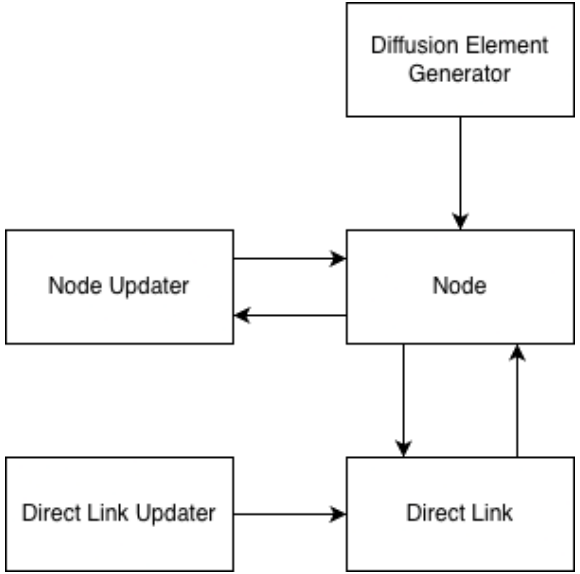
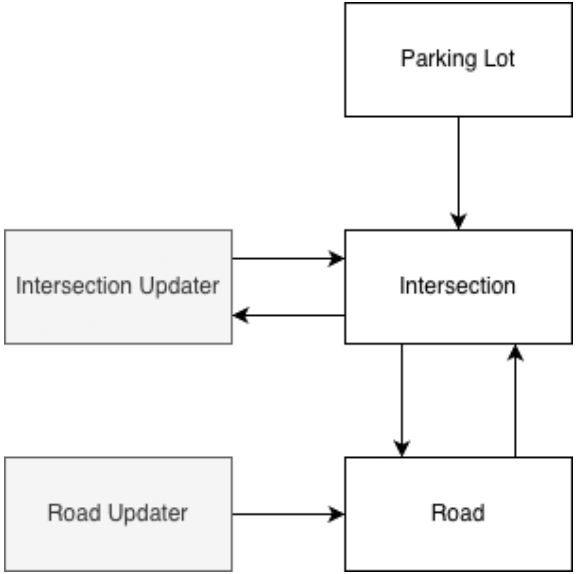
High-level System Architecture



Model Architecture

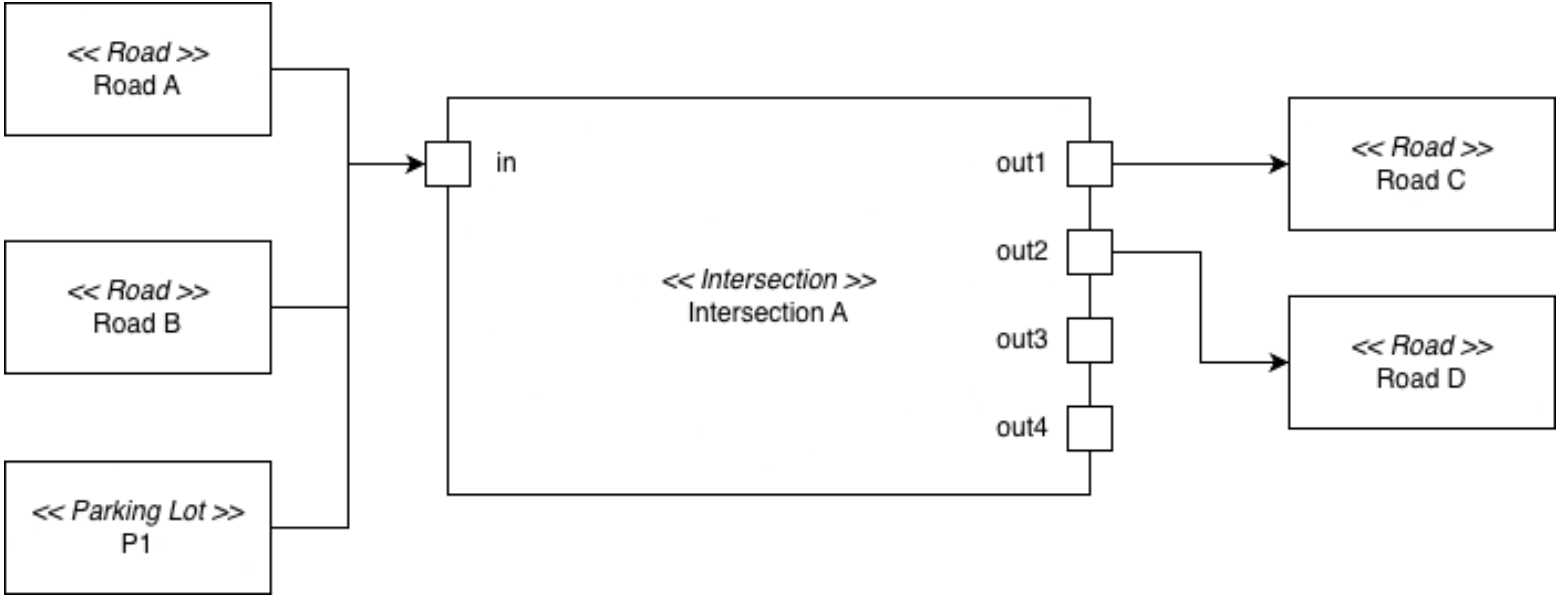


Diffusion Abstract Model

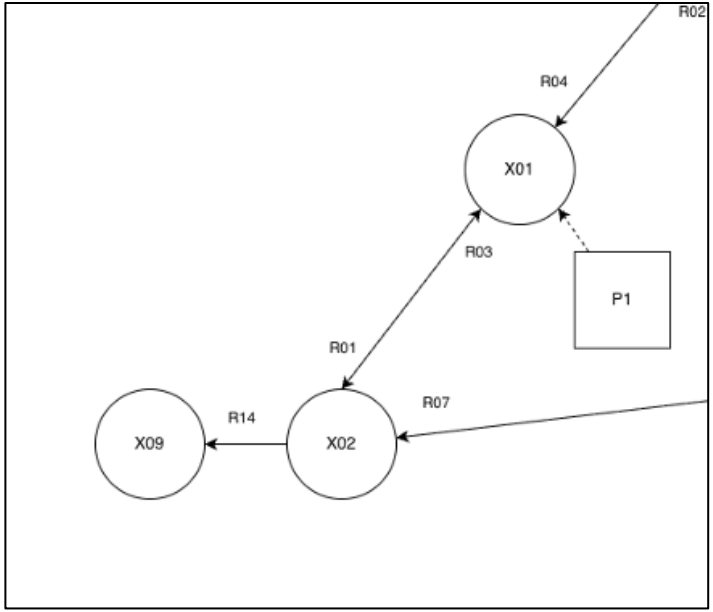


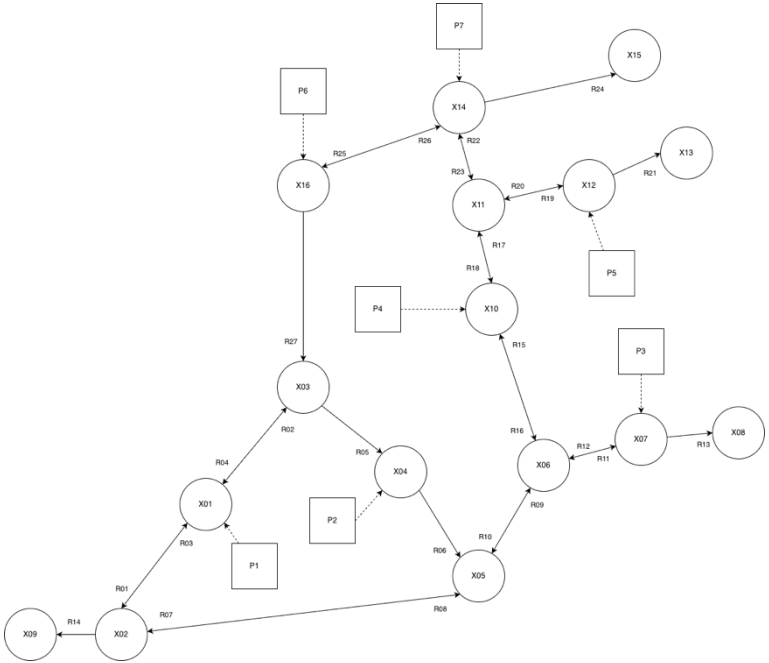
Implementation

DEVS Atomic Models



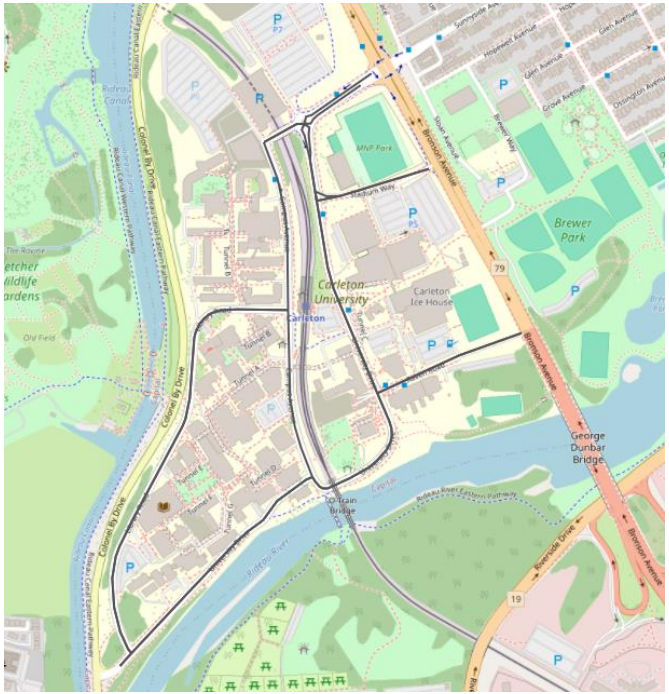
DEVS Coupled Model: University Campus





Configuration: Real-world Data

- Data collected from **OpenStreetMap**
  - Road length
  - Speed limit
  - Network structure (work in progress)



Configuration: OD data and Parking Lot Schedules

- OD Data
  - Student run **survey** determining flow rate
  - If at P1 and told to evacuate by Vehicle, which route would you take?
- Parking Lot Schedules
  - Synthetic
  - Periodically generates Vehicles, e.g., every 5 seconds
  - Capacity of parking lot

Simulation Output

```
root@main:~/cadmium-projects/model-campus-evacuation# cat output_data/raw/scenario_01_log.csv | grep "id=1,"
0,17,P7,exit,Vehicle{src=P7,id=1,dest=}
0,46,Roundabout,out2,Vehicle{src=P7,id=1,dest=Roundabout to Bronson Ave & University Dr}
14.69,12,Roundabout to Bronson Ave & University Dr,exit,Vehicle{src=P7,id=1,dest=Roundabout to Bronson Ave & University Dr}
```

- L1: At time = 0, Vehicle exits **P7** and enters **Roundabout**
- L2: At time = 0, **Roundabout** routes Vehicle to **Bronson Ave & Uni Dr**
- L3: At time = 14.69, Vehicle enters **Bronson Ave & Uni Dr** (evacuated)

Simulation Output

```
root@main:~/cadmium-projects/model-campus-evacuation# cat output_data/raw/scenario_01_log.csv | grep "id=2,"
0,19,P4,exit,Vehicle{src=P4,id=2,dest=}
0,22,P4 & University Dr,out1,Vehicle{src=P4,id=2,dest=P4 & University Dr to Raven Rd & University Dr}
20.27,35,P4 & University Dr to Raven Rd & University Dr,exit,Vehicle{src=P4,id=2,dest=P4 & University Dr to Raven Rd & University Dr}
20.27,14,Raven Rd & University Dr,out1,Vehicle{src=P4,id=2,dest=Raven Rd & University Dr to Campus Ave & University Dr}
84.71,9,Raven Rd & University Dr to Campus Ave & University Dr,exit,Vehicle{src=P4,id=2,dest=Raven Rd & University Dr to Campus Ave & University Dr}
84.71,36,Campus Ave & University Dr,out1,Vehicle{src=P4,id=2,dest=Campus Ave & University Dr to Library Rd & University Dr}
531.74,13,Campus Ave & University Dr to Library Rd & University Dr,exit,Vehicle{src=P4,id=2,dest=Campus Ave & University Dr to Library Rd & University Dr}
531.74,38,Library Rd & University Dr,out3,Vehicle{src=P4,id=2,dest=Library Rd & University Dr to Colonel By Dr & University Dr}
535.82,6,Library Rd & University Dr to Colonel By Dr & University Dr,exit,Vehicle{src=P4,id=2,dest=Library Rd & University Dr to Colonel By Dr & University Dr}
```

- L1: At time = 0, Vehicle exits **P4** and enters **P4 & Uni Dr**
- L2: At time = 0, **P4 & Uni Dr** routes Vehicle to **Raven Rd & Uni Dr**
- L3: At time = 20.27, Vehicle enters **Raven Rd & Uni Dr**
- ...
- L9: At time = 535.82, Vehicle enters **Colonel By Dr & Uni Dr** (evacuated)

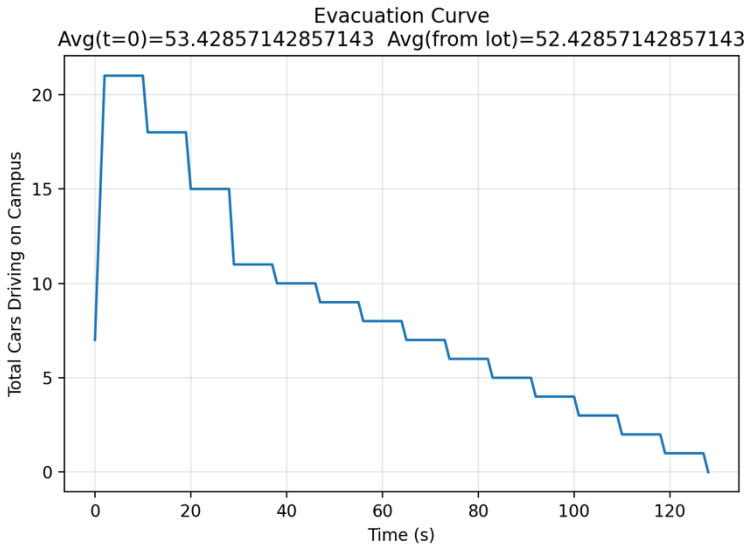
Testing

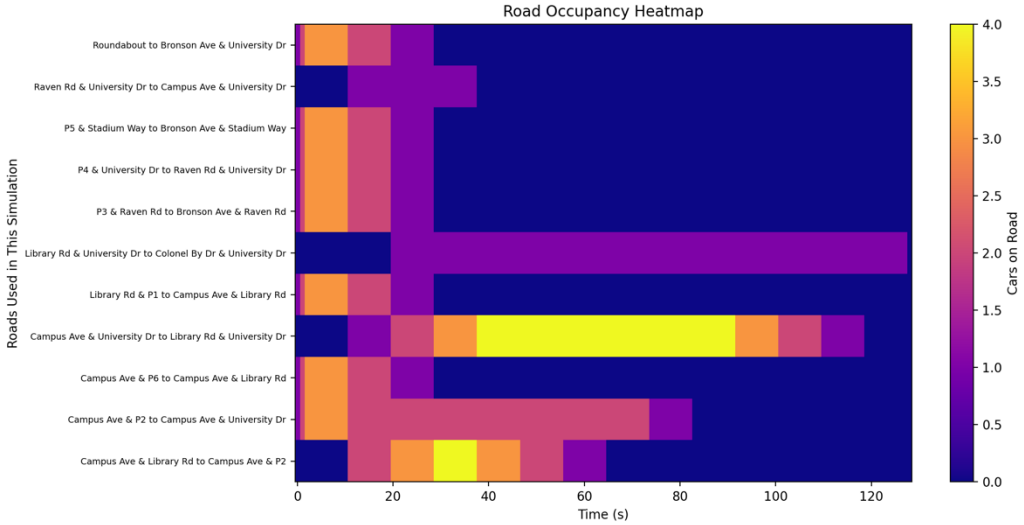
- **Manual verification** that all vehicles evacuate and all nodes are **reachable**
- **Assert statements** to check for **illegal states**
- **Unit tests** for atomic models (work in progress)

Running Experiments

- Testing for **quickest overall evacuation** while **minimizing congestion**
- **KPIs include**
  - Overall performance
  - Bottlenecks and congestions
  - Throughput

Visualization and Analysis





Conclusion

- Next Steps
  - **February 2:** Complete OD data collection
  - **February 16:** Complete tests & experiments
  - **March 2:** Complete visualization and analysis

- Deadlines
  - **March 20:** Final Report Draft
  - **March 27:** Poster Fair
  - **April 8:** Final Report

Thank You

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